oscillation cycle.

5 (a)(i) When one polaroid is rotated, resultant intensity of bright fringes decreases but the resultant intensity of dark fringes increases. The pattern becomes less defined as the contrast between the maxima and minima decreases. B1

When the axes of polaroids are perpendicular to each other, the fringe pattern disappears to give uniform illumination. B1

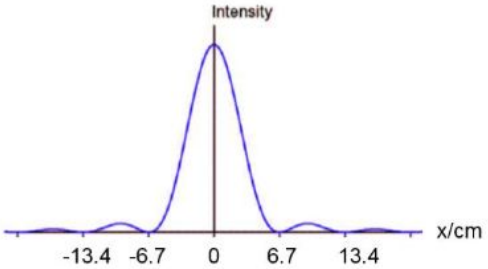
(a)(ii) 1. $\sin\theta = \frac{\lambda}{b} = \frac{644\times10^{-9}}{2.4\times10^{-5}}$ C1

$\Rightarrow \theta = 1.54^{\circ}$

$\tan 1.54^{\circ} = x/2.5 \Rightarrow x = 0.067$ m C1

Width = $2x = 2(0.067) = 0.134$ m A1

(a)(ii)
2.



- At least 3 maxima, symmetrical about O
- Correct shape, with decreasing intensity
- Correct spacing
- Correct label (Any missing above - minus 1 mark)

B3

(b)(i) According to Rayleigh criterion, when the central maximum of one image falls on the first minimum of another image, the images are said to be just resolved. B1

(b)(ii) Since 2.0° is greater than the minimum angle of 1.54° for the 2 images to be revolved, patterns resolved. B1

6a	$\theta = \frac{CV}{}$	